

Part A: Multiple choice

-Be sure that you have 6 different questions.

-Complete the following table by putting (the correct answer letter) in the table.

Q1) The solution of the differential equation $y' = -\frac{3x+y}{x} + 2$ is

a) $y = ce^x + 3x + 1$

b) $y = ce^x - 3x^2 + 1$

c) $y = ce^{2x} + 3x + 1$

d) $y = c + 3x + 1$

Q2) The solution of the IVP $y'' + 3y' + 2y = 0$, $y(0) = 1$, $y'(0) = 2$ is

a) $y = 4e^{-x} - 3e^{-2x}$

b) $y = 4e^{-x} + 5e^{-2x}$

c) $y = 3e^x - 4e^{-2x}$

d) $y = 6e^{-x} - 4e^{4x}$

Q3) The solution of the differential equation $y'' + 6y' + 9y = 0$ is

a) $y = c_1 e^{4x} + c_2 x e^{2x}$

b) $y = c_1 e^{-2x} + c_2 x e^{-2x}$

c) $y = c_1 e^{-3x} + c_2 x e^{-3x}$

d) $y = c_1 e^{3x} + c_2 x e^{3x}$

Q4) If the roots of characteristic equation of a homogeneous linear differential equation with constant coefficients are $\lambda = -2 \pm 4i$, the differential equation is

a) $y'' + 4y' + 13y = 0$

b) $y'' - 4y' - 20y = 0$

c) $y'' + 4y' - 20y = 0$

d) $y'' + 4y' + 20y = 0$

Q5) Find the second solution for $x^2 y'' - 2xy' - 4y = 0$ such that $y = x^4$

a) $y_2 = \frac{1}{5x}$

b) $y_2 = \frac{-1}{5x}$

c) $y_2 = \frac{5}{x}$

d) $y_2 = \frac{x}{5}$

Q6) Let $y = e^{ax}$ solution of the differential equation $y'' - 4y' + 4y = 0$ find a

a) -1

b) 1

c) 2

d) 3

Part B: Fill the Blanks (final answer)

Q7: The solution of the differential equation $y' - xy = 4e^{\frac{1}{2}x^2} \sqrt{y}$ is

$$\frac{2x}{\sqrt{y}} \sim \frac{4x e^{\frac{1}{2}x^2}}{\sqrt{y}}$$

Q8: The solution of the differential equation $y'' + 2y' + 2y = 0$ is

$$e^{-x} (-2 \pm \sqrt{-4})$$

Q9: The solution of the differential equation $(ye^{xy} - 2y^3)dx + (xe^{xy} - 6xy^2 - 2y)dy = 0$ isQ10: The solution of the differential equation $xy' + 2xy + 3y = x^{-2}e^{-3x}$ is

$$\frac{-2x}{e^{-2x} \cdot x}$$

Part C: Written Questions

(7 / marks)

Question 11: The general solution of differential equation is $y = x^{-3}(c_1 \cos(4 \ln x) + c_2 \sin(4 \ln x))$ Question 12: Solve $x^2 y'' - 2xy' - 4y = 0$



1- If $y' + p(x)y = q(x)$, then the general solution will be of the form $y =$

- a) $e^{\int (p(x)-q(x))dx}$ b) $e^{\int (q(x)-p(x))dx}$ c) $\int (q(x)-p(x))dx + c$ d) none

2- One of the following is a second order nonlinear D.E.

- a) $\sin(x)y'' + yy' = 1$ b) $\ln(y)y' + y = x$ c) $xy'' + y' = ye^x$ d) none

3- The value of m that makes $y = x^m$ a solution to $x^2y' - 3xy = 0$ is

- a) $-\frac{3}{2}$ b) 2 c) -3 d) 3

4- The particular solution of $(2xy - \sec^2 x)dx + (x^2 + 2y)dy = 0$, $y(0) = 2$, is $y =$

- a) $x^2y - \tan x + y^2 = 4$ b) $x^2y + \tan x + y^2 = 4$ c) $x^2y - \tan x - y^2 = 4$ d) none

5- If the integrating factor of $f(y)dx + (x + y^2)dy = 0$ is $\mu = y$, then $f(y) =$

- a) $\frac{y^2}{2} + g(x)$ b) $\frac{y}{2} + \frac{g(x)}{y}$ c) $\frac{y}{2} + \frac{c}{y}$ d) $\frac{y^2}{2} + c$

6- The general solution of the exact D.E. $(2x + y \cos(xy))dx + x \cos(xy)dy = 0$ is

- a) $\sin(xy) - x^2 = c$ b) $\sin(xy) + x^2 = c$ c) $\sin(xy) + 2x^2 = c$ d) none

7- The general solution of $(1 + 3y^2)y' = y \cos x$ is

- a) $\ln y + y^3 = \sin x + c$ b) $\ln y + y^3 = -\sin x + c$ c) $\ln y + y^2 = \sin x + c$ d) none

8- If the general solution of a D.E. is $y = c_1x^{-1} + x^{-2}(c_2 \cos(3 \ln x) + c_3 \sin(3 \ln x))$, then the D.E. is

- a) $x^3y'' + 8x^2y' + 23xy' + 13y = 0$ b) $x^3y'' + 8x^2y' - 23xy' + 13y = 0$
c) $x^3y'' - 8x^2y' + 23xy' + 13y = 0$ d) $x^3y'' - 8x^2y' - 23xy' + 13y = 0$

9- To solve the nonlinear D.E. $y' - 2x^2y = 3x^4y^{-2}$, $y > 0$, we convert it to L.D.E., then the integrating factor for the resulting Linear D.E. is $\mu =$

- a) $e^{-\frac{2}{3}x^3}$ b) e^{-3x^2} c) $e^{\frac{2}{3}x^3}$ d) e^{-2x^2}

10- Suppose $u = \frac{y}{x}$ then the general solution of $y' = \frac{y}{x} + e^{\frac{y}{x}}$ is $\ln x =$

- a) $e^{u^2} + c$ b) $e^{-u} + c$ c) $-e^u + c$ d) $-e^2 + c$

11- If $y_1 = x + 1$ is a solution of $(x^2 + 2x + 1)y'' + (x + 1)y' + Ay = 0$, then $y_2 =$

- a) $\frac{x-1}{2(x+1)}$ b) $\frac{-1}{4(x+1)^2}$ c) $x^2 + x$ d) $(x+1)\ln(x+1)$

12- The general solution of $y^{(4)} - 2y''' = 0$ is $y =$

- a) $c_1 + c_2x + c_3x^2 + c_4e^{2x}$ b) $c_1 + c_2x + c_3e^{2x}$ c) $c_1 + c_2x + c_3e^{-2x}$ d) none

13- Which of the following is a solution to $xy' - y = 2x^3$

- a) x^2 b) x^3 c) e^x d) x

GOOD LUCK.

maks reviewed by

1	2	3	4	5	6	7
B	A	D	A	B	B	A
8	9	10	11	12	13	
A	B+D	C	D	A	B	



Q4 (2 points): The solution of the I.V.P $y' = y$; $y(0) = 3$ is $y =$
A) $3e^x$ B) $e^x + 3$ C) $e^x + 2$ D) $2e^x$ E) None

Q5 (2 points): Which of the following DE's is of Bernoulli-type:
A) $xy' + yy' = 2\sqrt{y}$ B) $xy' + y'' = 2\sqrt{y}$ C) $xy' + y = 2\sqrt{y}$
D) $xy' + y^2 = 2\sqrt{y}$ E) None

Q6 (2 points): Determine whether the DE is Exact or not: $\sin(x) dx - ydy = 0$

Q7 (4 points): Solve the I.V.P. $y' - y = 5$; $y(0) = 0$